Mega8 Simulation Analysis Report

0. Executive summary

Mega8 is the **first full balance sweep on the re-axed champion roster** (T.36b, merged to `main` via PR #45) and the deepest to date: nine team sizes (2v2–10v10) × six weathers, 360 pieces (120 bases × 3 levels), with a combined median of **2,340 matches/piece** — roughly 10× the per-cell depth of mega7. Because the roster shape is otherwise unchanged, the mega7→mega8 deltas **isolate the re-axis**.

Five headline findings:

1. **Faction balance is sound.** Champion vs enemy win rate sits at 0.508 / 0.494 pooled and stays within ±0.015 at every team size. Every residual below is *within* the roster, not a champion/enemy asymmetry.

2. **The mage deficit is fixed.** This was the standing pathology in mega6/mega7 — mages were the floor role with a sharp small-format within-tier deficit (+0.079 at 2v2). The re-axis closes it: the within-tier mage deficit flips to **−0.011 at 2v2** (a −0.089 swing), and mage pooled `wr_delta` is now **+0.015** — a mild *surplus*. Mage win rate (0.522) now sits comfortably mid-pack.

3. **Role residuals are tight; apparent role spread is power-position, not mistuning.** Pooled win rates span support 0.459 to spellblade 0.565 (∼10pp), but the pooled `wr_delta` band is narrow (spellslinger −0.017 to mage +0.015). support (0.459) and tank (0.476) read low on raw win rate but their `wr_delta` is ≈0 — they are low-power-concentrated roles losing about what their power predicts, *not* under-tuned kits.

4. **Apex L3 high-power pieces under-deliver vs their power budget — the one open issue.** The new power-vs-budget boxplot (§7) shows residuals flat on 0 through $P \leq 20$, then the top L3 band (P 32–64) sags to −0.04−−0.05. By piece: **Aurion** (clear T10 L3, −0.151, the single worst piece), **Maw of the Drowned** (−0.131), **Hierarch** (−0.128), **Stone Warden** (−0.124). They win on raw power but less than their power says they should. Milder than mega7's −0.09 apex sag, but the same structural shape — the candidate for a T.36c pass.

5. **Weather favor is modest and the metric is now clean.** Own-weather edge is **+0.013 own-vs-counter** and **+0.010 own-vs-clear**, positive for every affinity. The `counter_weather_wr` zero-fill artefact that overstated this in earlier sweeps is fully resolved — **0.0% of rows** now carry a spurious counter==0 (the buggy column-average and the raw computation agree: +0.012 vs +0.013).

Two caveats lifted since mega7. (a) **Timeouts no longer climb with team size** — they hold flat at ∼9–11% across all sizes (mega7 hit 44% at 10v10), so large-format role reads are now trustworthy. (b) **Per-cell depth is no longer thin** (median 2,340 matches/piece), so per-stage win-rate spread is real signal rather than binomial noise.

Three supplementary lenses (§§10–12). A roster-health index (§10) scores the sweep against the win-rate-band framework Riot uses for League of Legends: **84% of pieces are within ±0.05 of budget and only 2.8% (10 pieces) beyond ±0.10** — a healthy roster by that standard. A combat-pacing pass (§11) and an unsupervised behavioral-archetype clustering (§12) then re-frame the apex sag: **it is a *pacing* problem.** High-power pieces split into fast finishers and slow grinders, and the under-delivery lives almost entirely in the *slow durable* apex archetype (wins by grinding toward the tick cap), not the fast finishers.

1. Dataset and method

Field	Value
Directory	`results/mega/mega8/`
Run date	2026-06-18
Engine state	current working tree after the T.36b champion re-axis (PR #45 merged): roster stat/playstyle rebalance
Stages	2v2, 3v3, 4v4, 5v5, 6v6, 7v7, 8v8, 9v9, 10v10 (no 1v1)
Per-stage rated rows	19,440
Combined pieces	360 (120 bases × 3 levels)
Weathers	clear, cloudy, mist, rain, snow, thunder
Median matches/piece (combined)	2,340
max-ticks	12,000 (engine default — sudden death engaged)
Baseline	mega7 (`cache_mega7.rds`)

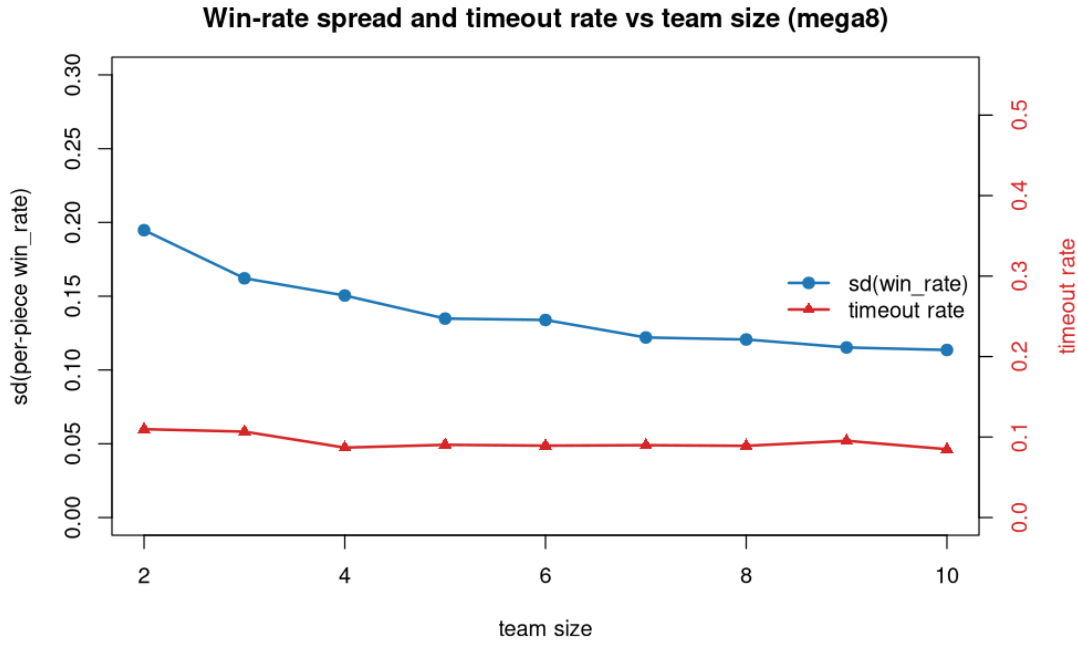
The combined file is the canonical aggregate. `ratings_combined.csv` pools every battle (pair-game weighted) across all stages and weathers, so its per-piece win rates and cross-weather metrics are the reliable read. All §7–§9 outlier work uses it.

`wr_delta` **interpretation.** `expected_wr` is the deterministic power-threshold model: 1.0 if your team's ΣP exceeds the opponent's, 0.0 if less, 0.5 if equal, averaged over the actual opponent field. `wr_delta = win_rate − expected_wr`. Near 0 = on-budget; negative = kit under-delivers vs power; positive = over-delivers. `expected_power` is the T.18 scalar $P(T, L) = 2^{(T-1)/3+\text{triplings}(L)}$, triplings = {L1:0, L2:1, L3:3}.

2. Aggregate balance

stage	champion	enemy	sd(win_rate)	timeout
2v2	0.507	0.496	0.195	0.110
3v3	0.510	0.491	0.162	0.107
4v4	0.509	0.492	0.150	0.087
5v5	0.508	0.493	0.135	0.090
6v6	0.505	0.498	0.134	0.089
7v7	0.514	0.489	0.122	0.090
8v8	0.506	0.496	0.121	0.089
9v9	0.507	0.495	0.115	0.095
10v10	0.505	0.495	0.114	0.085

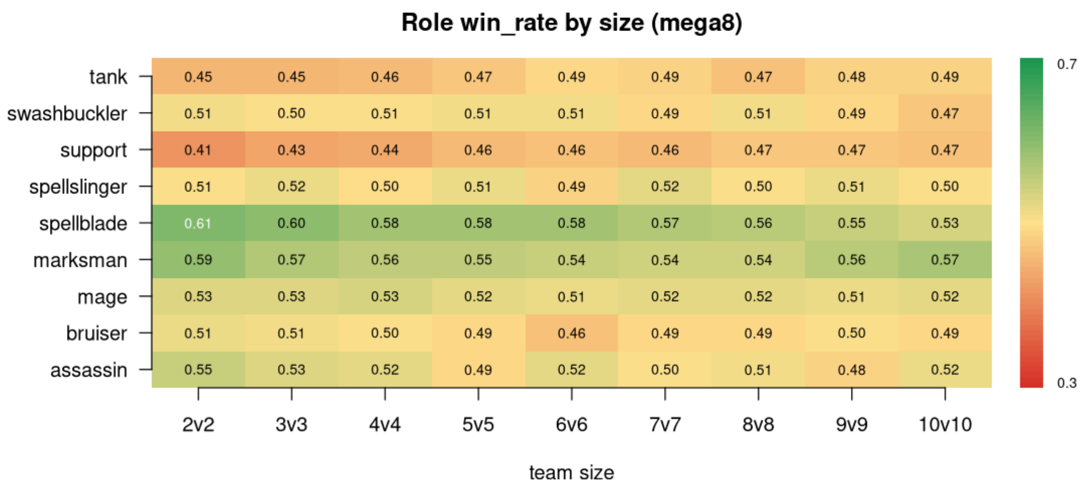
Champion/enemy parity holds across all formats (±0.015). Win-rate spread declines smoothly with team size (0.195 → 0.114) as larger teams average individual variance out — and because per-cell depth is now high, this is a *real* effect, not the noise-inflated reading mega7 carried. **Timeouts stay flat at ∼9–11% across all sizes** — mega7's large-format timeout blow-up (44% at 10v10) is gone, so every size resolves organically and role reads are comparable end-to-end.



win_rate spread and timeout vs team size

3. Role balance

3.1 Win rate by team size



Role win_rate by size (heatmap)

Pooled (combined) role win rates:

role	win_rate
support	0.459
tank	0.476
bruiser	0.492
swashbuckler	0.497
spellslinger	0.507
assassin	0.510
mage	0.522
marksman	0.557
spellblade	0.565

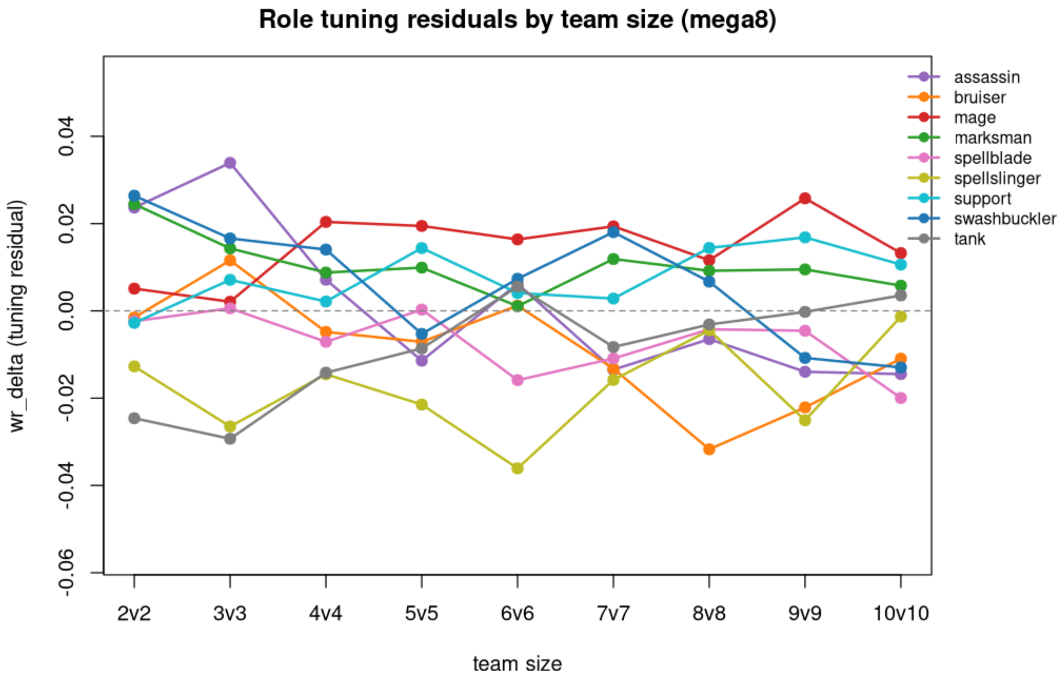
Spellblade and marksman are the ceiling; support and tank the floor. But raw win rate conflates kit tuning with where a role sits on the power curve — see §3.2.

3.2 Tuning residuals (wr_delta)

Pooled (combined) role `wr_delta`:

role	wr_delta
spellslinger	−0.017
bruiser	−0.011
spellblade	−0.009
tank	−0.006
assassin	−0.003
swashbuckler	+0.004
support	+0.008
marksman	+0.009
mage	+0.015

The residual band is narrow (±0.017) — the roster is well-tuned against power. The story splits from raw win rate: **support (0.459 raw) and tank (0.476 raw) carry ≈0** `wr_delta` — they lose about what their (low) power predicts, so the low win rate is power-position, not mistuning. **spellblade has the second-highest raw win rate but a *negative* residual (−0.009)** — it wins on power, and front-loads: 0.61 at 2v2 decaying to 0.53 at 10v10 (small-team snowball). marksman and mage genuinely over-deliver on kit (+0.009/+0.015).



Role tuning residuals by team size

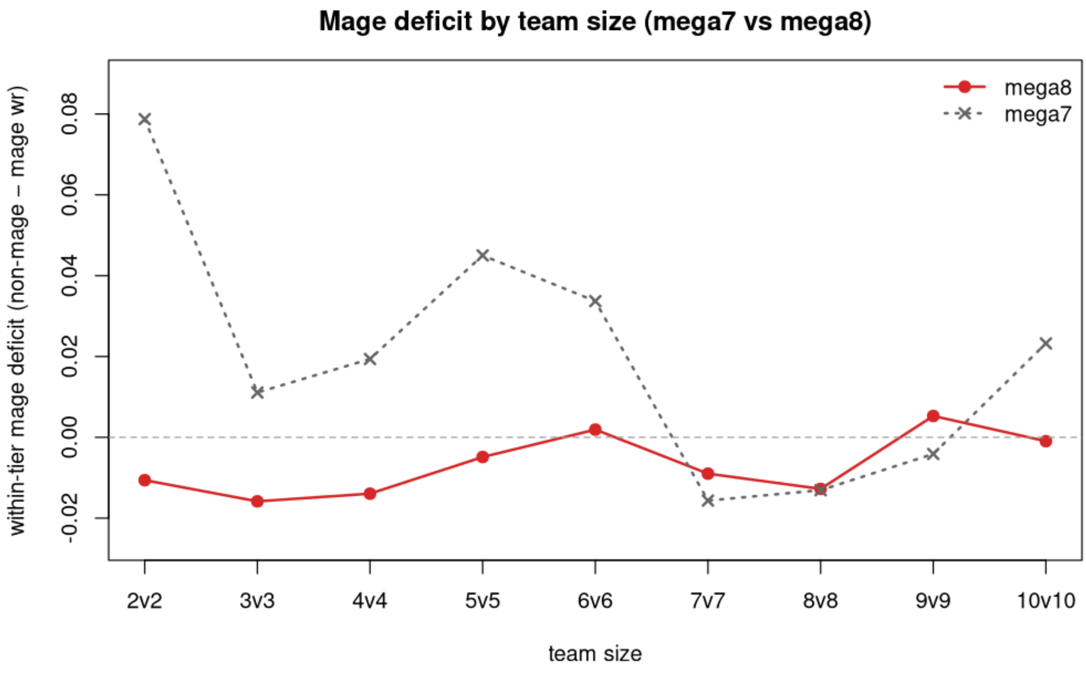
4. Mage deficit — closed

stage	mage	nonmage	within_tier_def	cor(tier,wr)
2v2	0.529	0.497	−0.011	0.555
3v3	0.527	0.497	−0.016	0.538
4v4	0.535	0.495	−0.014	0.499
5v5	0.523	0.498	−0.005	0.487
6v6	0.513	0.500	+0.002	0.430
7v7	0.522	0.499	−0.009	0.462
8v8	0.522	0.498	−0.013	0.460
9v9	0.513	0.499	+0.005	0.406
10v10	0.519	0.498	−0.001	0.457

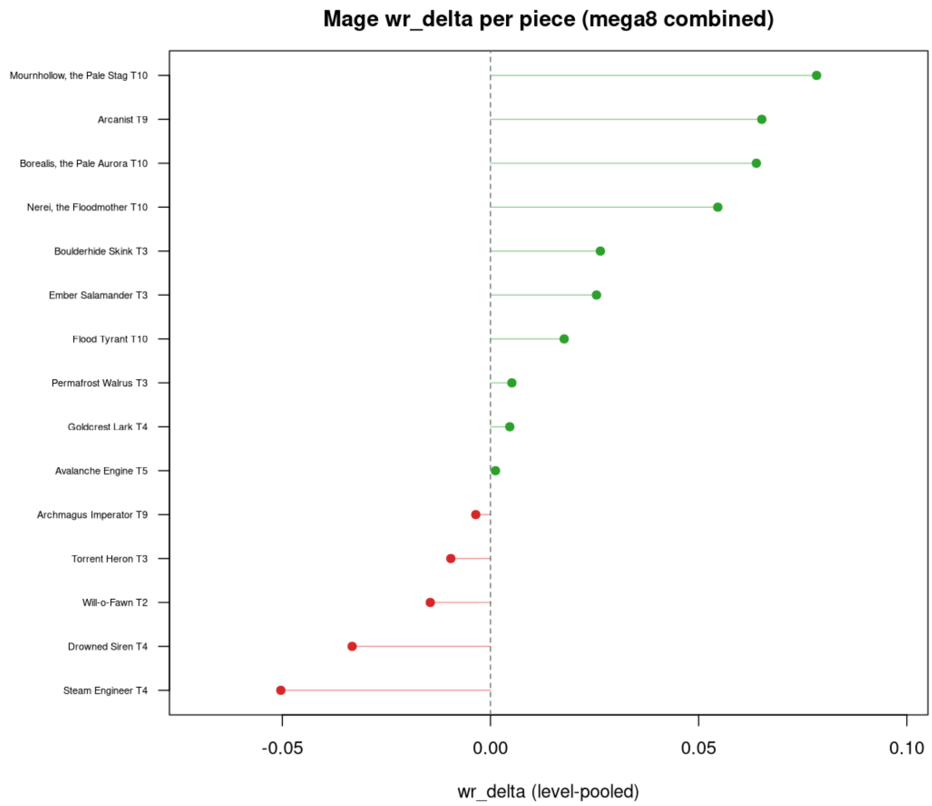
The within-tier mage deficit is gone at every size — mages now sit at or slightly above their tier cohort throughout. Against mega7 at matched sizes the swing is decisive at small formats:

stage	mega7 deficit	mega8 deficit	Δ
2v2	+0.079	−0.011	−0.089
3v3	+0.011	−0.016	−0.027
4v4	+0.019	−0.014	−0.033
5v5	+0.045	−0.005	−0.050
6v6	+0.034	+0.002	−0.032
10v10	+0.023	−0.001	−0.024

(Positive = mages behind their tier; a negative Δ = improvement.) The re-axis targeted exactly the small-format weakness and landed it. A handful of individual mages still read under power (Steam Engineer T4 L3 −0.114, Drowned Siren T4 L3 −0.079) but these are now ordinary tail pieces, not a role-wide floor.



Mage deficit by team size: mega7 vs mega8

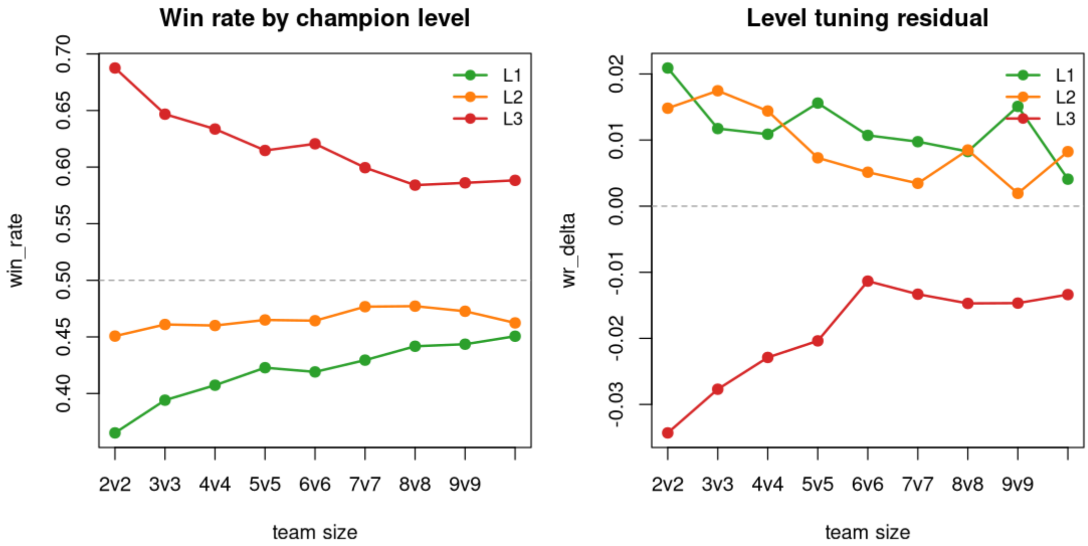


Mage wr_delta per piece (combined)

5. Level effects

level	win_rate	wr_delta	timeout
1	0.427	+0.011	0.092
2	0.468	+0.008	0.095
3	0.608	−0.017	0.098

The level premium is steep (L1→L3 = +0.18 raw) but `wr_delta` stays small: L1/L2 slightly over-deliver, **L3 slightly *under*-delivers (−0.017)**. That L3 sag is the level-axis shadow of the apex-power issue in §7 — the highest-power state is where kits stop fully converting power into wins. By design otherwise; no broad action needed.



Win rate and tuning residual by champion level

6. Weather effects

Own-weather advantage is real, positive for every affinity, and **modest (∼+0.01)**. The `counter_weather_wr` zero-fill artefact that overstated weather favor in earlier sweeps is now fully resolved: **counter==0 fires in 0.0% of rows**, so the buggy column-average (+0.012) and the correct raw computation (+0.013) agree.

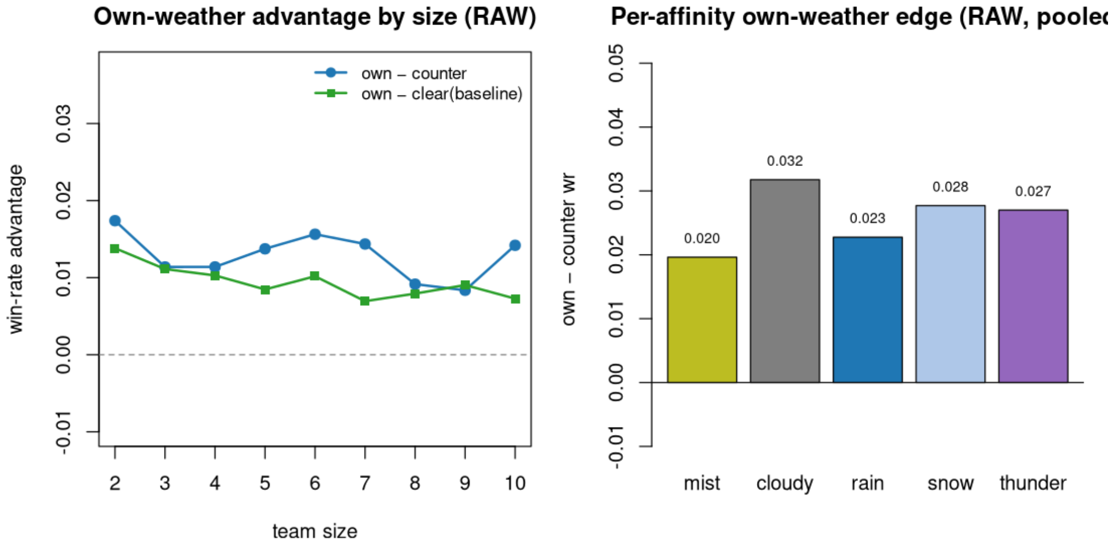
Own-weather advantage by size (own = win rate in own affinity weather; clear = no-favor baseline; counter = weather that preys on you):

stage	clear (base)	own	counter	own − counter	own − clear
2v2	0.500	0.514	0.496	+0.017	+0.014
4v4	0.500	0.510	0.499	+0.011	+0.010
6v6	0.500	0.510	0.495	+0.016	+0.010
8v8	0.500	0.508	0.499	+0.009	+0.008
10v10	0.500	0.507	0.493	+0.014	+0.007

Per-affinity (raw, pooled all stages):

affinity	clear	own	counter	own − clear	own − counter
mist	0.521	0.526	0.506	+0.005	+0.020
cloudy	0.505	0.524	0.492	+0.019	+0.032
rain	0.500	0.519	0.496	+0.018	+0.023
snow	0.516	0.527	0.499	+0.011	+0.028
thunder	0.497	0.517	0.490	+0.020	+0.027

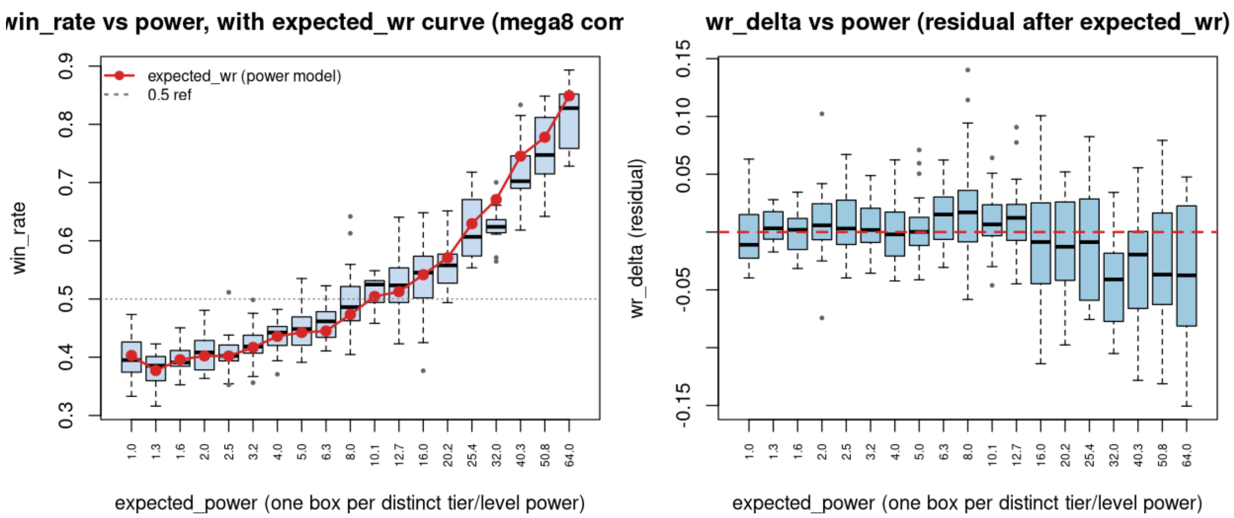
Every affinity gains in its own weather and loses (or is flat) in its counter — the favor system works as designed. The magnitude is single-digit win-rate points, so weather is a flavour-and-edge axis, not a dominant one. Sensitivity (per-piece spread of per-weather win rate) rises gently with team size (0.056 at 2v2 → 0.079 at 10v10), as more pieces share the same weather field.



Weather: own-advantage by size and per-affinity edge

7. Power vs budget — the apex sag

This sweep adds a per-piece view of win rate against the power scalar that drives `expected_wr`. Because mega8 draws teams at random, high-power pieces routinely meet low-power ones, so **win_rate *should* climb with power** — that climb is the deterministic power model working, not a finding. The diagnostic is the *residual*: `wr_delta` after the power curve is removed should sit flat on 0 at every power level. Each box is one distinct `expected_power`; the 30 (tier,level) combinations collapse to 19 distinct powers because $P(T, L)$ interleaves tier and level (e.g. T3/L1 = T1/L2 = 2.0).



Left: win_rate per power bin with the expected_wr curve overlaid (boxes track the curve). Right: wr_delta residual per power bin (flat on 0 = tuned).

Reading. The left panel's boxes hug the red `expected_wr` curve — the power model predicts win rate well across the whole range. The right panel is flat on 0 through $P \leq 20$ (residuals slightly positive, ∼+0.01–0.02, through the mid-band), then **the top L3 band (P = 32, 40, 51, 64) sags negative (−0.04 to −0.05) with wide boxes**. That is the apex sag: the highest-power pieces under-convert their raw power into wins. It is milder than mega7 (−0.09) but structurally identical, and it lines up with the L3 residual in §5 — the issue is the *apex L3 state*, not any one tier.

Named fliers (1.5×IQR). R draws boxplot outliers as unlabelled dots; these are the pieces deviating most from their *power cohort* (others at the same P), so they are the prime within-power tuning candidates. A flier is mechanically a piece beyond 1.5×IQR of its cohort; the *read* column says what that deviation likely means. Recurring patterns (tags used in all four flier tables):

AG — apex grinder: high-power L3 in a durable/slow kit; wins on raw stats but stalls toward the tick cap, so it under-converts power (§11–12). *Fix: closing/burst, not stats.* **EB — early bloomer:** low-tier (T1–3) kit stronger than its power band; over-delivers. *Fix: trim early scaling.* **DA — dead apex:** an L3 piece that also loses on *raw* win rate — kit barely converts at all. *Fix: damage/survivability.* **CH — cohort split:** win_rate far from same-power peers (a role×power mismatch). **LF — level-flip:** over-budget at one level, under at another (mis-shaped level scaling).

win_rate panel — deviates from cohort win rate at the same power:

P	**piece** (role)	**wr**	**why a flier / what it could mean**
2.5	Frostplate Tortoise T5/L1 (tank)	0.352	CH-low — a tank at low power loses more than its peers; durability buys little when base stats are small.
2.5	Veldt Pronghorn T2/L2 (tank)	0.511	CH-high — over-performs the low-P field; cheap kit punches up.
3.2	Hollowed Wisp T3/L2 (assassin)	0.356	CH-low — fragile assassin folds against the same-P cohort before it can burst.
3.2	Iron-Collared Hound T3/L2 (swash)	0.498	CH-high — wins more than its P3.2 peers.
4.0	Drowned Siren T4/L2 (mage)	0.371	CH-low / early under — mid mage under-delivers (also a residual flier).
8.0	Borealis T10/L1 (mage)	0.613	CH-high — a T10 king already beats its P8 peers at L1; king kit front-loaded.
8.0	Springfrog T1/L3 (support)	0.642	EB — a T1 piece winning 64% at P8 is over-scaled for its tier.
16	Steam Engineer T4/L3 (mage)	0.377	DA — an L3 mage that still loses 62% of games; kit not converting.
32	Mirewarden Toad T7/L3 (tank)	0.565	AG (CH-low) — apex tank grinds; low for the P32 cohort.
32	Marshghast Boar T7/L3 (bruiser)	0.571	AG (CH-low) — apex bruiser stalls.
32	Thunder Bull T7/L3 (support)	0.700	CH-high — unusually strong apex support for P32.
40	Glade Heron T8/L3 (marksman)	0.833	CH-high — top apex finisher; wins big but on-budget (fast, not a grinder).

wr_delta panel — residual after the power curve is removed:

P	**piece** (role)	Δ	**why a flier / what it could mean**
2.0	Conscript T1/L2 (swash)	−0.074	Early under — weak starter kit, loses even after adjusting for low power.
2.0	Springfrog T1/L2 (support)	+0.102	EB — over-budget at T1 already.
5.0	Coral Colossus T5/L2 (tank)	+0.051	Over — tank punching above its P5 budget.
5.0	Hierarch T8/L1 (support)	+0.059	LF — Hierarch is *over* at L1 but the worst apex *under* at L3: level scaling mis-shaped.
5.0	Caged Banshee T5/L2 (support)	+0.071	Over — mid support above budget.
8.0	Borealis T10/L1 (mage)	+0.114	Over — king kit already over-budget at L1 (cf. its win_rate flier above).
8.0	Springfrog T1/L3 (support)	+0.140	EB — largest positive residual in the sweep; the clearest over-scaled low-tier kit.
10	Powder Sapper T2/L3 (support)	−0.046	Under — a low-tier L3 support that lags its cohort.
10	Veilfang Wolf T8/L2 (swash)	+0.064	Over — swashbuckler above budget.
13	Ember Salamander T3/L3 (mage)	+0.077	EB — low-tier mage over its budget.
13	Arcanist T9/L2 (mage)	+0.091	Over — high-tier mage above budget (a healthy over-performer).

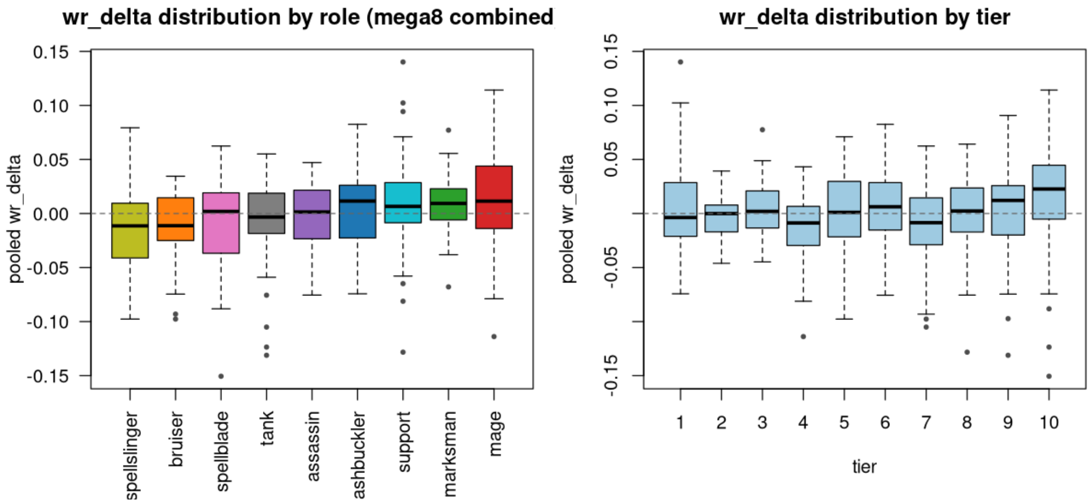
8. Timeouts

role	**2v2**	**5v5**	**8v8**	**10v10**
assassin	0.047	0.078	0.076	0.091
bruiser	0.142	0.100	0.104	0.097
mage	0.056	0.074	0.081	0.067
marksman	0.048	0.061	0.065	0.070
support	0.162	0.100	0.094	0.087
tank	0.191	0.126	0.109	0.103

The mega7 timeout blow-up is gone. Timeouts no longer climb with team size — they peak at small formats for the durable low-burst roles (tank 0.19, support 0.16, bruiser 0.14 at 2v2, where two tanky pieces can grind past the cap) and fall as team size rises. At no size does any role approach mega7's 40%+ stalls. Large-format role reads are now trustworthy.

9. Outliers and tuning priorities

9.1 Distribution shape



wr_delta distribution by role and by tier (boxplots)

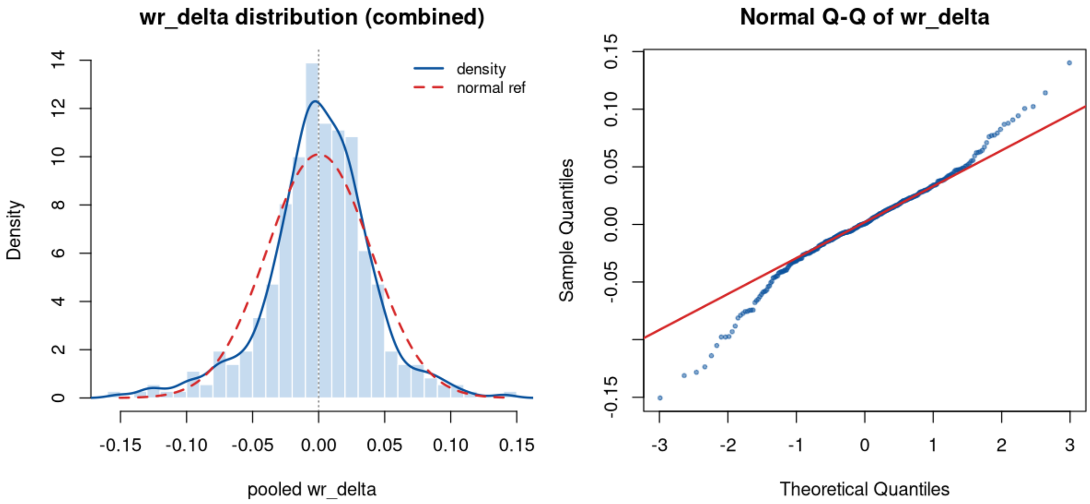
Named fliers (1.5×IQR) for both boxplots (tags as defined in §7). The unlabelled dots above, recovered — the pieces most out of line with their *role* cohort and their *tier* cohort. These are the same names that dominate §9.2, now attributed to the cohort that flags them.

by role — residual vs others in the same role:

role	piece	Δ	why a flier / what it could mean
bruiser	Dredge-Hulk T7/L3	−0.098	AG — apex bruiser; stalls toward the cap, under-converts power.
bruiser	Marshghast Boar T7/L3	−0.093	AG — second apex bruiser grinder; the role's L3 ceiling drags.
mage	Steam Engineer T4/L3	−0.114	DA — the worst mage; loses on raw win rate too, not just budget.
marksman	Voltaic Diviner T5/L3	−0.068	Under — apex marksman lagging an otherwise over-budget role.
marksman	Umbra T10/L1	+0.077	Over — king over-budget at L1; pulls the marksman box up.
spellblade	Aurion T10/L3	−0.151	AG — the single worst piece in the sweep; apex king that grinds.
support	Hierarch T8/L3	−0.128	AG — its on-death barrier helps allies, not itself; apex support stalls.
support	Coppercrest Stork T4/L3	−0.081	Under — mid support lagging its role.
support	Hoarfrost Owl T4/L3	−0.065	Under — mid support lagging its role.
support	Lostlight Wisp T1/L3	+0.094	EB — over-scaled T1 support.
support	Springfrog T1/L2	+0.102	EB — over-scaled T1 support.
support	Springfrog T1/L3	+0.140	EB — biggest over-performer; T1 kit far above its band.
tank	Maw of the Drowned T9/L3	−0.131	AG — durable apex tank; wins by grinding, under-converts.
tank	Stone Warden T10/L3	−0.124	AG — apex tank grinder.
tank	Mirewarden Toad T7/L3	−0.105	AG — apex tank grinder.
tank	Inquisitor T6/L3	−0.076	AG — milder apex tank grinder.

by tier — residual vs others in the same tier:

tier	**piece** (role)	Δ	**why a flier / what it could mean**
1	Springfrog (support)	+0.140	EB — T1 kit over-scaled; the strongest piece relative to its tier.
3	Ember Salamander (mage)	+0.077	EB — low-tier mage above its tier cohort.
4	Steam Engineer (mage)	−0.114	DA — breaks below its T4 cohort and loses raw; kit under-built.
7	Mirewarden Toad (tank)	−0.105	AG — apex grinder, worst of its tier.
7	Dredge-Hulk (bruiser)	−0.098	AG — apex grinder, drags T7.
8	Hierarch (support)	−0.128	AG — apex grinder; worst of T8.
9	Maw of the Drowned (tank)	−0.131	AG — apex grinder; worst of T9.
9	Cliffeyrie Eagle (sslinger)	−0.097	AG — spellslinger that stalls at apex.
10	Aurion (spellblade)	−0.151	AG — apex king; worst of T10 and the sweep.
10	Stone Warden (tank)	−0.124	AG — apex tank king, drags T10.
10	Aerion (spellblade)	−0.088	AG — second apex spellblade king grinder.

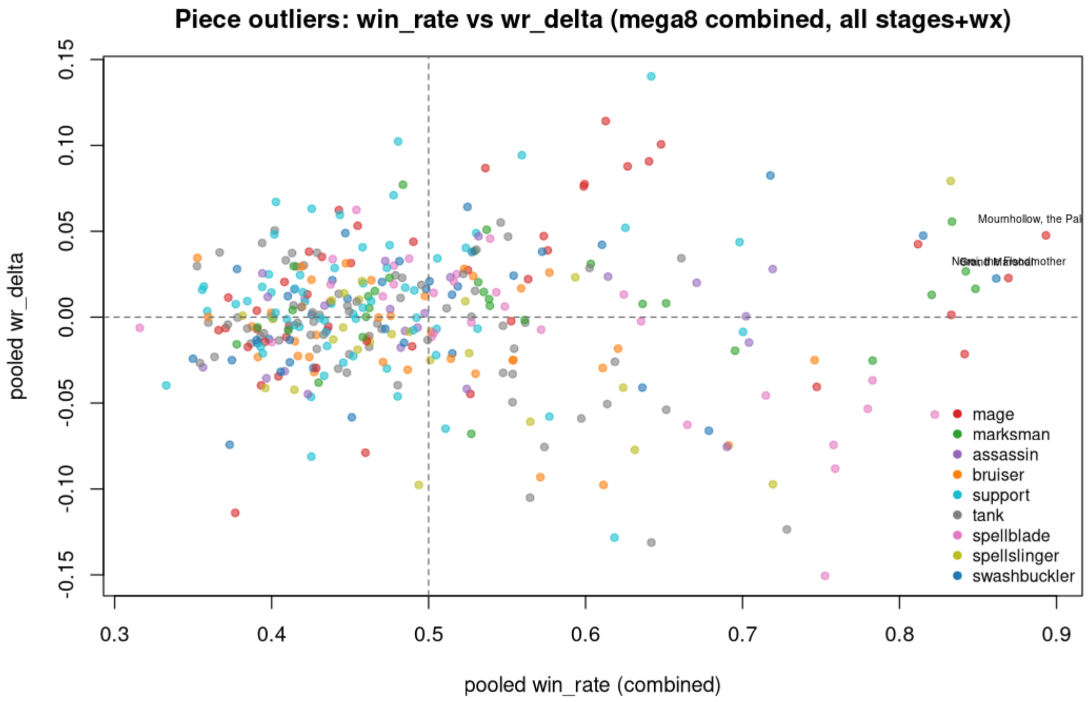


wr_delta distribution: histogram + density vs normal, and normal Q-Q

The pooled `wr_delta` distribution is near-normal in the body and only mildly heavy-tailed — most pieces sit within ±0.08 of budget. By tier the boxes are tight and centred near 0 except the **T10 box, which skews negative** (the apex sag of §7). The named outliers below are genuine within-tier tail events (`wd_z` column), not a fat-tailed mess.

9.2 Absolute outliers (largest |wr_delta|)

`wd_z` = within-tier z-score of `wr_delta`. $|z| \gtrsim 2 \approx$ a genuine per-tier outlier.



Piece outliers: win_rate vs wr_delta (combined)

Most under-tuned (pooled `wr_delta`):

name	affinity	role	tier	level	win_rate	wr_delta	wd_z
Aurion, the First Dawn	clear	spellblade	10	3	0.753	−0.151	−2.81
Maw of the Drowned	rain	tank	9	3	0.642	−0.131	−2.95
Hierarch	clear	support	8	3	0.618	−0.128	−2.96
Stone Warden	cloudy	tank	10	3	0.728	−0.124	−2.35
Steam Engineer	clear	mage	4	3	0.377	−0.114	−3.00
Mirewarden Toad	rain	tank	7	3	0.565	−0.105	−2.47
Battlemage	clear	spellslinger	5	3	0.494	−0.098	−2.78
Dredge-Hulk	rain	bruiser	7	3	0.611	−0.098	−2.27
Cliffeyrie Eagle	cloudy	spellslinger	9	3	0.719	−0.097	−2.20
Marshghast Boar	mist	bruiser	7	3	0.571	−0.093	−2.15

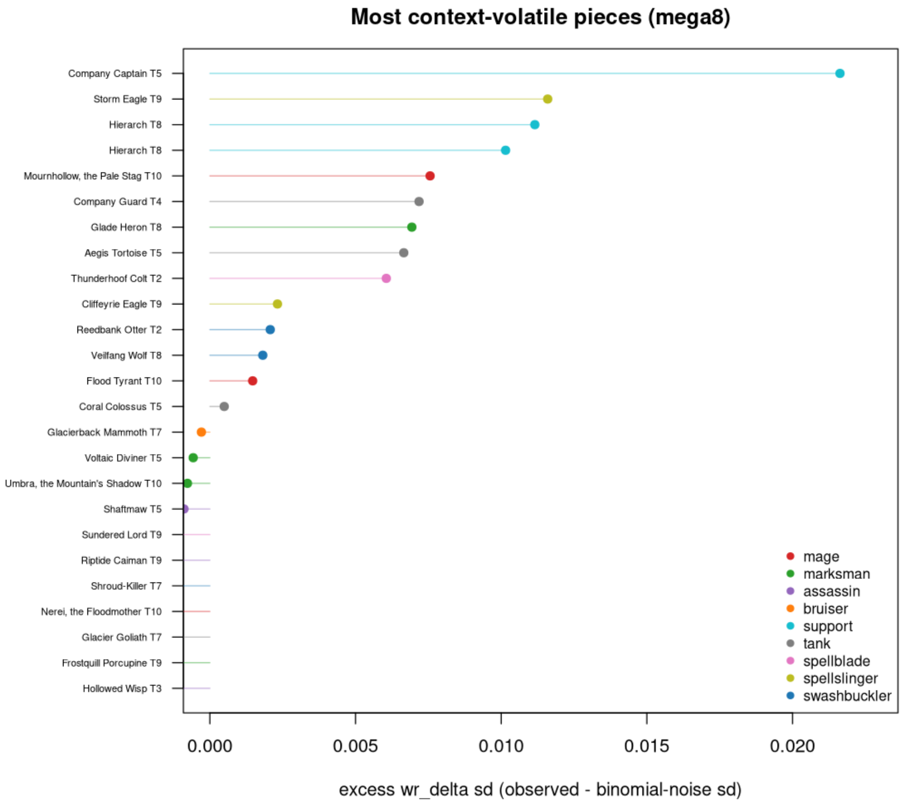
Most over-tuned:

name	affinity	role	tier	level	win_rate	wr_delta	wd_z
Springfrog	rain	support	1	3	0.642	+0.140	+2.91
Borealis, the Pale Aurora	snow	mage	10	1	0.613	+0.114	+1.72
Springfrog	rain	support	1	2	0.481	+0.102	+2.08
Mournhollow, the Pale Stag	mist	mage	10	2	0.648	+0.101	+1.49
Lostlight Wisp	mist	support	1	3	0.559	+0.094	+1.90
Arcanist	clear	mage	9	2	0.640	+0.091	+1.98
Iceclaw Lynx	snow	swashbuckler	6	3	0.718	+0.083	+2.37
Ember Salamander	clear	mage	3	3	0.599	+0.078	+2.87

The under-tuned list is the apex sag made concrete: all but the lowest-tier entries are **T7–10 L3** pieces that win on power but miss budget — led by the four T8–10 apex pieces **Aurion, Maw of the Drowned, Hierarch, Stone Warden**. The over-tuned list is led by a T1 support (Springfrog, z=+2.9 at L3 — the single oddest piece, an early-game over-performer) and high-tier mages winning slightly above budget. Full 30-row table: `tables/m8_abs_outliers.csv`.

9.3 Spread outliers — context-volatility of wr_delta

Each piece's match-weighted `sd(wr_delta)` across its 54 contexts, minus the expected binomial-noise sd, ranked by the excess. A near-0 pooled residual that hides large per-context swings is its own concern.



Most context-volatile pieces (excess wr_delta sd beyond noise)

Spread is almost entirely sampling noise. Only **14 of 360** pieces show `excess_sd > 0`, and the excesses are small (≤0.022):

name	affinity	role	tier	wd_pooled	wd_sd	excess_sd
Company Captain	clear	support	5	−0.062	0.105	+0.022
Storm Eagle	thunder	spellslinger	9	+0.059	0.066	+0.012
Hierarch	clear	support	8	−0.164	0.093	+0.011
Mournhollow, the Pale Stag	mist	mage	10	+0.042	0.051	+0.008
Company Guard	clear	tank	4	−0.062	0.080	+0.007
Glade Heron	rain	marksman	8	+0.046	0.062	+0.007
Aegis Tortoise	clear	tank	5	−0.029	0.088	+0.007

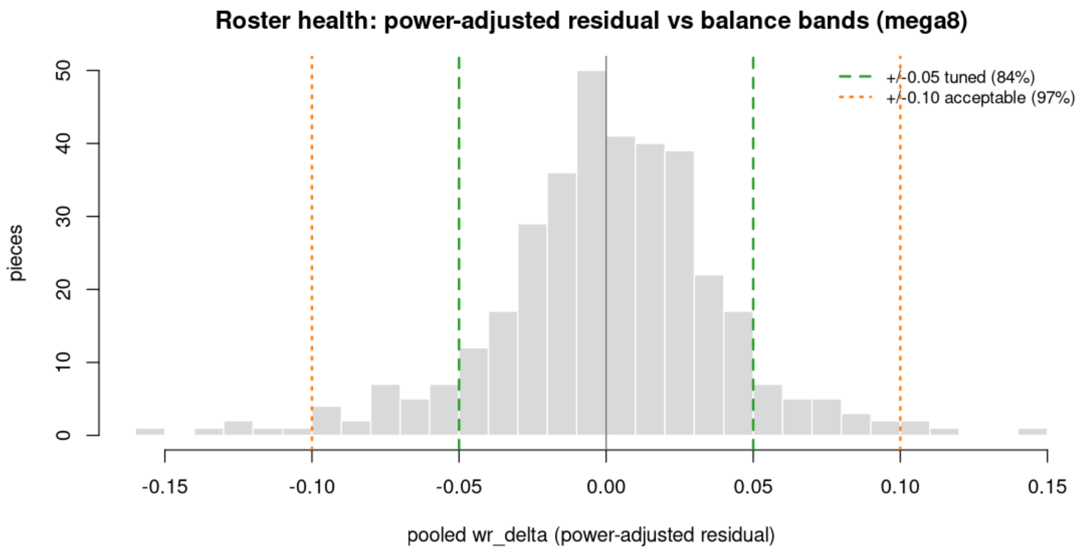
Takeaway: `wr_delta` is not meaningfully context-volatile in mega8 — the outliers worth acting on are the *absolute* ones in §9.2. Hierarch appears on both lists (under-tuned *and* slightly volatile), reinforcing it as a priority. Full table: `tables/m8_spread_outliers.csv`.

10. Roster health index

Beyond naming outliers, it is worth scoring the *whole* roster with a single health read. The games industry's standard is the **win-rate band**: a balanced unit should sit near 50%, and deviation past a threshold flags it for a change. Riot's published champion-balance framework treats ~48/52% as the weak/strong band and ~45/55% as "broken" for League of Legends.[1] Two adaptations are needed here. First, raw win rate is power-confounded in a random-draw sweep, so the band is applied to `wr_delta` (the power-adjusted residual), where 0 is the target. Second, the sim has no pick/ban, so there is no "Presence" signal — but it gains something human telemetry never has: **every piece is exposed equally** (median 2,340 matches), so there is no popularity bias to correct for.

metric	value
within ±0.05 `wr_delta` (tuned)	84.2%
within ±0.10 (acceptable)	97.2%
beyond ±0.10 (outlier)	2.8% (10 pieces)
sd(`wr_delta`)	0.039
sd(win_rate)	0.114
Gini(win_rate)	0.121

The roster is healthy by this standard. Five sixths of all pieces are tightly on-budget and the residual sd (0.039) is a quarter of the raw win-rate sd (0.114) — i.e. most of the win-rate spread is power doing its job, not mistuning. The low Gini (0.121) of win rate confirms outcomes are not concentrated in a small clique of dominant pieces. The 10 pieces beyond ±0.10 are the §9.2 list, headed by the apex four.



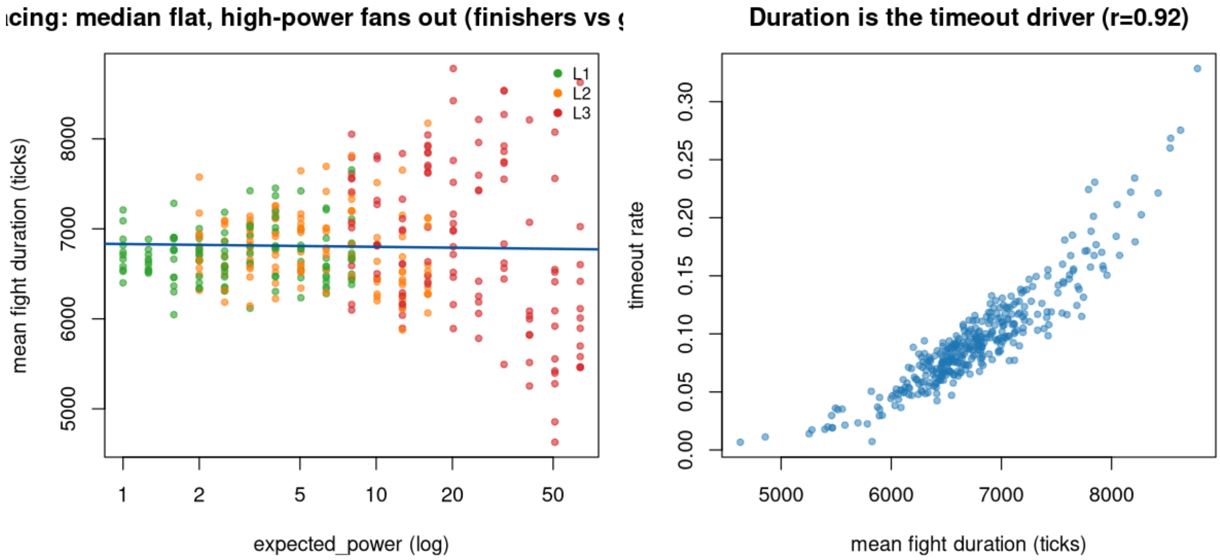
Power-adjusted residual distribution against the tuned (±0.05) and acceptable (±0.10) bands

11. Combat pacing

The combined file carries `mean_duration_ticks` per piece, never used in prior reports. Median fight duration is **6,710 ticks** — 56% of the 12,000-tick cap — so the typical battle resolves with comfortable headroom, which is why timeouts are low (§8). Duration is the direct timeout driver (`cor(duration, timeout)=0.92`, right panel).

[1] Riot Games, */dev: Champion Balance Framework*, https://nexus.leagueoflegends.com/en-us/2019/05/dev-champion-balance-framework/.

The pacing–power relationship is the interesting one. The *median* duration is essentially flat across power (L1 6,711, L2 6,782, L3 6,816 ticks — a 1.5% spread), but the **variance fans out sharply at high power**: low/mid-power pieces cluster tightly around 6,700 ticks, while high-power L3 pieces spread across the *entire* 4,600–8,800-tick range. The apex does not resolve at one speed — it **splits into fast finishers (close in <5,500 ticks) and slow grinders (drag past 8,000)**. The weak overall `cor(power, duration)=−0.19` is the fast-finisher tail pulling the mean down, not a uniform speed-up.



Left: duration vs power, coloured by level (median flat, high-power variance fans out). Right: duration drives timeout.

12. Behavioral archetypes (clustering beyond designer roles)

Win rate alone hides *how* a piece wins. A growing balance-research line argues for clustering units on their *behavioral* signature rather than reading per-unit win rates in isolation.[2] Following that approach, we ran an unsupervised **k-means clustering** (k=6) over six standardized outcome features per piece — win_rate, `wr_delta`, timeout_rate, weather_sensitivity, duration, and power — with *no* knowledge of the designer role labels, then asked which roles fall into which cluster.

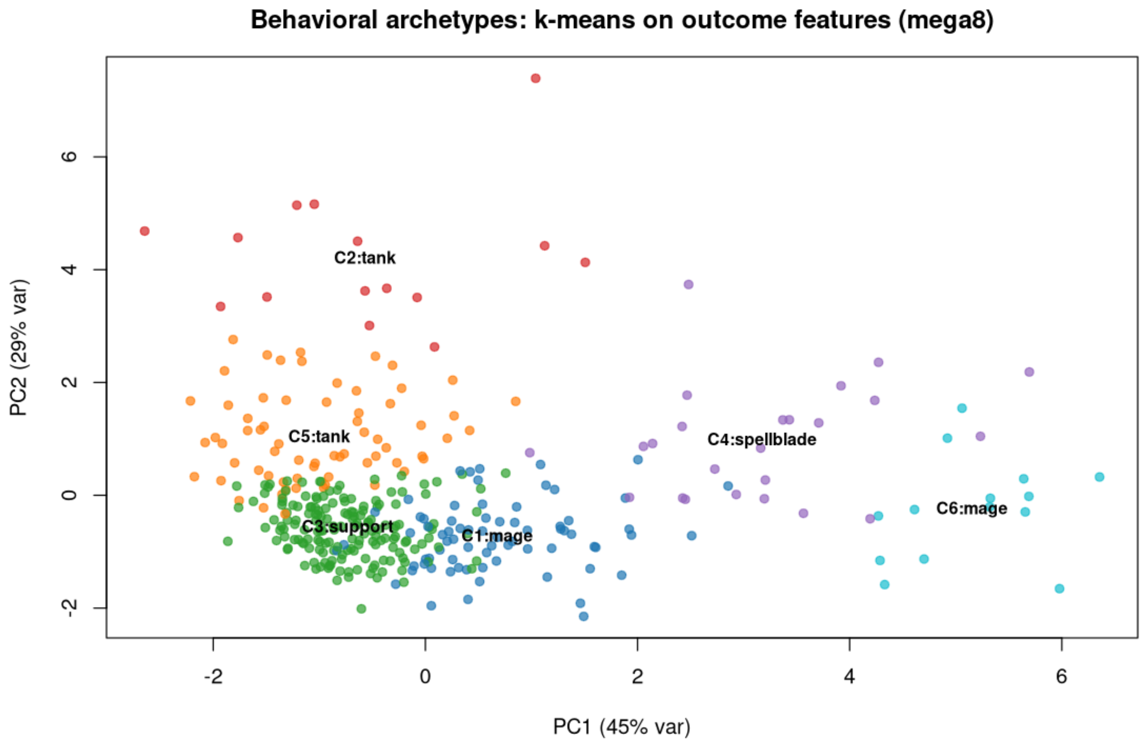
cluster	n	win_rate	wr_delta	timeout	power	top role	purity
C1 (efficient mid)	73	0.535	+0.033	0.071	11.4	mage	0.23
C2 (durable grinder)	15	0.607	−0.069	0.209	33.7	tank	0.67
C3 (low-power base)	168	0.423	−0.005	0.089	4.3	support	0.22
C4 (high-power carry)	24	0.701	−0.051	0.053	44.7	spellblade	0.38
C5 (midline wall)	66	0.497	+0.005	0.140	10.8	tank	0.47
C6 (apex finisher)	14	0.824	+0.033	0.022	53.1	mage	0.36

Finding 1 — behavior is organised by power and durability, not by designer role. Cluster purity is low (0.22–0.67): every designer role smears across multiple clusters. The two principal components of the feature space are **power/strength (PC1, 45% of variance)** and **durability/pace (PC2, 29%)** — so a piece's outcome signature is set far more by *where it sits on the power curve and how long its fights run* than by its role tag. Roles are a design vocabulary; they are not behaviorally separable in outcome space.

Finding 2 — the apex sag is a pacing split, and this is the actionable result. The high-power pieces divide into two clusters that win very differently: **C6 "apex finisher"** (power 53, timeout 0.02, fast, `wr_delta` *+0.033*) and **C2 "durable grinder"** (power 34, timeout 0.21, slow, `wr_delta` *−0.069*). The under-delivery of §7 is concentrated almost entirely in C2 (and the high-power C4 carries, −0.051) — pieces that *grind* toward

[2]See e.g. *Beyond Win Rates: A Clustering-Based Approach to Character Balance Analysis in Team-Based Games*, arXiv:2502.01250; and *Identifying and Clustering Counter Relationships of Team Compositions in PvP Games*, arXiv:2408.17180.

the tick cap — while the pieces that close fast are on or above budget. So the apex four (Aurion, Maw of the Drowned, Hierarch, Stone Warden) under-deliver **because their kits stall the fight, not because their raw numbers are low.** A T.36c fix should add closing/burst pressure, not flat stats.



k-means archetypes in outcome-feature PCA space; centroids labelled by dominant designer role. PC1 = power, PC2 = durability/pace.

13. Methodology: mapping to industry balance analytics

How studios balance from data, and where this sweep sits.

Win-rate bands (Riot). The dominant practitioner method: assume a balanced unit wins ~50%, flag deviation past a band, patch on a cadence. Used here in power-adjusted form (§10).

Latent-strength models (Bradley–Terry / Elo). Estimate a scalar strength per unit from head-to-head results; the win probability of i over j is a logistic function of their strength difference.[a] Our `expected_wr` power-threshold model is a deterministic cousin; a true Bradley–Terry fit needs the 1v1 matrix (skipped in this sweep — a candidate for a dedicated 1v1 mega run). These scalar models are known to miss **non-transitive** (rock-paper-scissors) structure, which motivates the clustering line below.

Counter / archetype clustering. Recent work clusters compositions by counter-relationship or behavioral signature to expose structure win rates hide (§12).

What the sim has that telemetry lacks. No pick/ban means no "Presence" signal — but also **no popularity or skill confound**: every piece is played an equal, large number of times by an identical (deterministic) policy, so a low win rate is never "unpopular/misplayed," it is the kit. And the engine is byte-deterministic, so any finding re-runs exactly. The trade-off: no human meta, no skill-expression axis, and AI policy can under-use a kit a human would exploit — so treat these as *designer-intent* reads, not live-meta predictions.

[a] The Bradley–Terry paired-comparison model is the standard framework for latent strengths from pairwise results.

14. Recommendations

- **Apex L3 under-delivery (priority for T.36c) — fix the *pace*, not the stats.** The four T8–10 L3 pieces **Aurion, Maw of the Drowned, Hierarch, Stone Warden** under-deliver vs power (−0.12 to

−0.15). The clustering (§12) localises *why*: they belong to the slow "durable grinder" archetype that wins by dragging fights toward the tick cap, not the fast "apex finisher" archetype which is on-budget. So the fix is **closing pressure / burst / a finisher mechanic**, not more flat stats — adding raw power would only deepen the grind. This is the one structural residual left.

- **Hierarch specifically.** On both the under-tuned and the context-volatile lists — the clearest single target.
- **Mage deficit: fixed, no action.** The re-axis closed the standing mage problem (§4); mage is now a mild surplus role. Monitor that it does not drift further positive.
- **Weather favor: no action.** Working as designed at ∼+0.01; the overstating metric bug is fully resolved (0.0% zero-fill rows).
- **Watch the small-format over-performers** (Springfrog, Lostlight Wisp at T1 L3) — low-tier supports punching above budget. Minor, but the mirror image of the apex sag.

15. Reproducibility

```
# 1. Run the sweep (workers = CPU cores)
python -m tools.simulation.mega --workers 16 --skip 1v1 \
   --out results/mega/mega8

# 2. Analysis + tables + cache
Rscript reviews/mega_sim/13_mega8.R

# 3. Core figures (m8_01..m8_12)
Rscript reviews/mega_sim/14_mega8_plots.R

# 4. Supplementary analyses (health/pacing/archetypes, m8_13..m8_15)
Rscript reviews/mega_sim/15_mega8_extra.R

# 5. This PDF
xelatex -output-directory=reviews/mega_sim \
   reviews/mega_sim/mega8_analysis_report.tex
```

Tables written to `reviews/mega_sim/tables/m8_*.csv`; cache to `cache_mega8.rds`; figures to `reviews/mega_sim/plots/m8_*.png`. The engine is byte-deterministic — same tree + same flags reproduce the CSVs exactly.